Circular motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate inward net force to a change in velocity direction.

SUCCESS CRITERIA - I can...

1. Draw tangent velocity and inward acceleration.
2. Use v^2/r or a supplied acceleration.
3. Identify the real interaction providing centripetal force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “centripetal” and “radius” to explain your answer. Sentence starters are optional.

centripetal: directed toward the center of a circular path

radius: distance from the center to the edge of a circle or sphere

net force: the vector sum of all forces on one object

01 Warm-up: bring back an earlier idea

A projectile’s velocity is (**9**, **-12**) m/s.
1. **Find its speed**.
2. What information would locate it?

Word help: projectile = an object moving after launch under the stated forces

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

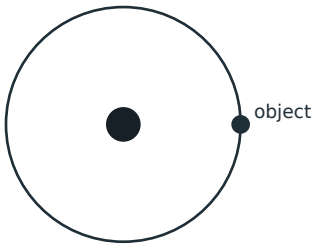
02 Worked example: explain the prediction

Reference: $a_c = v^2/r$; $F_{net} = mv^2/r$.
c means centripetal: toward the center. This is the net inward force, not an extra force.

Worked reasoning: A **2 kg** object moves at **3 m/s** in a circle of radius **3 m**. Inward acceleration is v^2/r=**3 m/s²**; net force is **6 N** inward.

Your turn: Draw tangent velocity and inward net force at two different points.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Circular motion

antocciphysics

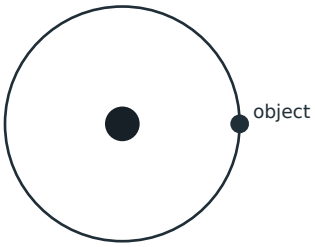
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

1. **Add velocity** and acceleration arrows to the marked point on the circle.
2. **Name one interaction** that could provide the inward force.

Sentence starter: In my work, I used … to show … . I labeled it … .



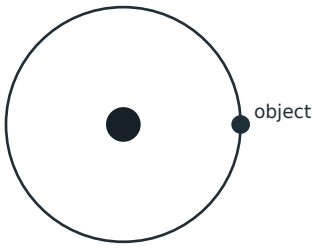
The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

A **0.50 kg** puck on a string moves at **4 m/s** in a horizontal circle of radius **2 m**. **Draw velocity** and net-force arrows. **Calculate inward acceleration** and net force. Then **predict both quantities** at twice the speed and the same radius.

Sentence starter: I predict … . I used … to make this prediction because … .

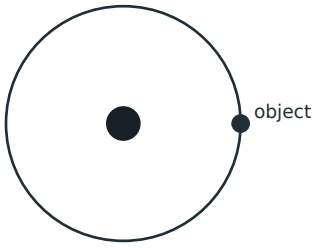


Marzano check - circle: ① ② ③

05 Explain your reasoning

A string breaks while a ball moves in a circle. Neglect other forces over the next instant. **Sketch its initial path after** release.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

Circular motion

antocciphysics

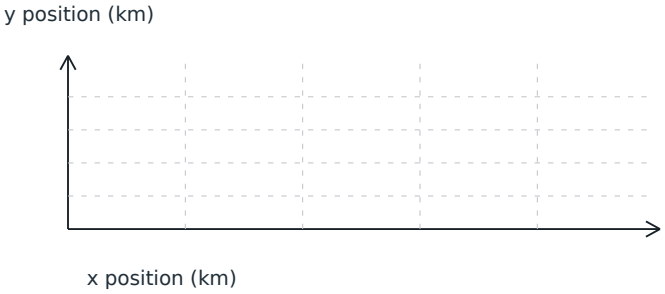
Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A simplified calculation predicts position (**-80,100**) km with bounds ±**8 km** in x and ±**10 km** in y. **Write a short briefing giving** both intervals and two reasons it is not a real asteroid collision forecast.
Label the given and predicted quantities on the diagram.

Sentence starter: I predict … . I used … to make this prediction because … .

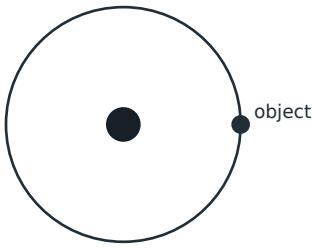


Marzano check - circle: ① ② ③

07 Exit check

At fixed speed and mass, double circle radius. How do acceleration and net force change?
1. Explain from equations.
2. **Label the radius** and **show velocity** and inward acceleration on the circle.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

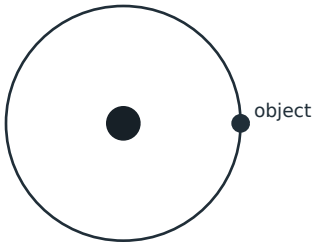


Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

The maximum horizontal force on a **0.5 kg** object is **8 N**. For a **1 m** circle, find the maximum speed. Then double the radius and **find the new limit**. **Label the radius** and **show velocity** and inward acceleration on the circle.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③